AMIRMOHAMMAD JAFARI

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GitHub

PROFESSIONAL EXPERIENCE

System Group — Software Engineer (Golang) Aug 2025 – Present

- *Fixed Asset Management (FAM):* Building depreciation calculation services so asset values over useful life are computed correctly and exposed to other platform services through shared APIs.
- *Payroll and severance (HCM):* Implemented payroll and severance calculation services in Go within a multi-service monorepo, using PostgreSQL and gRPC so employment rules produce consistent, auditable compensation results for financial reporting.
- *Payslip reporting (HCM):* Integrated JasperReports with payroll outputs to generate employee payslips, binding calculated figures into printable documents used in production payroll cycles.

Ariyana Shabake Afagh — AI Software Engineer (Python) Oct 2024 – Aug 2025

- *AI-Powered Assistant Development:* Built Baxii, an AI assistant for Instagram, WhatsApp, and Telegram, using RabbitMQ, GraphQL, gRPC, and MSSQL in a microservices architecture.
- *Service Design and Automation:* Designed Telegram services with Telethon and the OpenAI API, and deployed Instagram webhooks with Flask, enabling real-time task execution for 3,000+ users.
- *Backend Optimization:* Migrated assistant services from Django to gRPC, reducing API response time to 100 ms and improving scalability.

IPM — Research Intern Jun 2023 – Sep 2023

- *Emotion Captioning Models:* Implemented multilingual emotion captioning systems for the ICCV Emotion Challenge using the ArtELingo dataset.
- *Deep Learning Architectures:* Experimented with ResNet and Vision Transformers (ViT) using PyTorch, TensorFlow, and OpenCV to improve captioning performance.

EDUCATION

Amirkabir University of Technology *Sep 2020 - Apr 2025*
B.Sc. in Computer Engineering Overall GPA: 18.48/20

Allame Helli 2 High School *Sep 2016 - Sep 2020*
Diploma of Theoretical Mathematics and Physics Overall GPA: 19.85/20

RESEARCH EXPERIENCE

Research Assistant at Amirkabir University of Technology *Jan 2024 – Mar 2025*
Under supervision of Dr. Maryam AmirMazlaghani

- **Gradient-Based Adversarial Attacks on LLMs:** Designed and implemented a white-box framework for automated adversarial prompt generation using differentiable decoding (soft prompts, Gumbel Softmax, learnable token distributions) and backpropagation guided by a pretrained safety classifier. Developed as part of my B.Sc. thesis: *"Automated Gradient-Based Attack on Large Language Models"*.
- **Evaluation and Vulnerability Analysis:** Applied gradient-guided optimization inspired by the GBRT paper to adversarial prompt engineering, achieving an average 0.31 toxicity score (Perspective API) and demonstrating the susceptibility of LLMs to gradient-based prompt optimization.

TECHNICAL SKILLS

Languages: Go, Python, C/C++, Java

Backend: gRPC, protobuf, GORM, GraphQL, RabbitMQ, Flask, Django

Databases: PostgreSQL, MSSQL, MongoDB, SQL

ML / AI: PyTorch, TensorFlow, scikit-learn, OpenAI API

Tools & Platforms: Docker, Kubernetes, Linux, Git, JasperReports

NOTABLE PROJECTS

VGG16_CUDA
Developed a basic VGG16 implementation using CUDA for GPU acceleration, optimizing custom kernels without cuDNN. Despite its simplicity, it demonstrated a speedup of approximately 10 times compared to a similar project optimized with cuDNN, highlighting expertise in GPU programming and parallel processing.

IR_System
Developed a custom Persian-news search engine using normalized TF–IDF and cosine similarity over an inverted index with champion lists. The system indexed over 12,000 articles with positional postings and per-term IDF, and returned the top-10 ranked results in under 500 ms on average (single-machine). Implemented query-time cosine scoring with document-length normalization and an optional Jaccard baseline for comparison.

AI_Pac
Implemented search algorithms (DFS, BFS, UCS, A*) for maze navigation, adversarial search (minimax, alpha-beta pruning) for ghost–Pac-Man interaction and Q-learning with probabilistic inference (particle filters, Bayes' nets) for hidden ghost tracking. Achieved optimal paths on standard mazes with A* while reducing node expansions by nearly 40% compared to uninformed search, and trained Q-learning agents that reached competitive performance (average rewards approaching heuristic agents after a few thousand episodes).

NUMEX
Built a pure functional programming language interpreter in Racket, implementing 30+ language constructs (arithmetic, logic, recursion, pairs, records, macros) along with a type inference system. Extended the language with higher-order functions (foldr, concatenate, bind) and optimized closures via free-variable analysis (eval-exp-c).

HONORS AND AWARDS

- **National Countrywide University Entrance Exam (Konkour), Iran** *2020*
  Ranked in the top 0.1% among more than 150,000 participants.
- **Iran National Cognitive Neuroscience Competition** *2016*
  Participated in the Iran National Cognitive Neuroscience Competition, gaining familiarity with signal processing concepts using MATLAB and foundational neuroscience principles.